Opella Healthcare Italy PACKAGING TEAM	
Code:	938347
Update:	V03 - 30-07-2025
GMID code:	832802
Current item code:	845863
Product/Item type:	Istruzione Enterogermina 2 MLD - 20 Flac
Country:	Malaysia
Artwork by:	Origgio
Plant:	Origgio

For Opella use only

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Technical Data

Format:	185 x 210 mm.
Plant barcode:	1057
Number of colours:	1
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Fonts:	Futura Book / Book Oblique / Bold / Bold Oblique
Minimum point size of text:	7 pt
Layout of Cutting:	N.A.
Packaging Line:	EG 096

GDO - Graphic Department Origgio
Opella Healthcare Italy S.r.l. - Viale Europa, 11
21040 Origgio (VA) - Italy - Tel. +39 - 02 96 10 559

Vietata la manomissione - Rendere dopo la stampa

15 mm

Enterogermina®

2 billion/ 5 ml, oral suspension
Spores of poly-antibiotic resistant *Bacillus clausii*

QUALITATIVE AND QUANTITATIVE COMPOSITION
Oral suspension: 2 billion (strains SIN, O/C, T, N/R) For a full list see list of Excipients.

PRODUCT DESCRIPTION
Oral suspension: Whitish, opalescent, liquid.

CLINICAL PARTICULARS
Therapeutic Indications
Treatment and prophylaxis of intestinal dysmicrobism and subsequent endogenous dysvitaminosis. Therapy for aiding the recovery of the intestinal microbial flora, altered during the course of treatment with antibiotics or chemotherapeutic agents. Acute and chronic gastrointestinal disorders in breastfeeding infants, attributable to intoxication or intestinal dysmicrobism and dysvitaminosis.

Posology and method of administration
Adults: 2-3 minibottles per day; children: 1-2 minibottles per day; breastfeeding infants: 1-2 minibottles per day.
Minibottles: administration at regular intervals (3-4 hours), taking the contents of the minibottles as it is or diluting it in water or other drinks (e.g. milk, tea, orange juice).
This medicine is for ORAL use only. DO NOT inject or administer in any other way.

Contraindications
Hypersensitivity to the active ingredient or any of the excipients.

Special warnings and precautions for use
Special warnings:
Bacteraemia/sepsis
Post-marketing cases of bacteraemia, septicaemia and sepsis have been reported in patients being immunocompromised or severely ill, and in preterm infants. In some critically-ill patients, the outcome was fatal. ENTEROGERMINA should be avoided in these patient groups (see section Undesired effects)
This medicinal product is for oral use only. Do not inject or administer in any other way. Incorrect use of the medicinal product has caused severe anaphylactic reactions such as anaphylactic shock.

Precautions for use:
During antibiotic therapy, the product should be administered in the interval between one dose of antibiotic and the next.
Any presence of visible corpuscles in the minibottles of ENTEROGERMINA is due to aggregates of Bacillus clausii spores and does not, therefore, suggest that the product has been altered.
Shake the minibottle before use.

Interactions with other medicinal products and other forms of interaction
There are no known medicinal interactions subsequent to the concomitant administration of other drugs.

Fertility, Pregnancy and lactation
Pregnancy
No data available on the use of Enterogermina in pregnant women; therefore, no conclusions can be drawn regarding the safety of the use of Enterogermina during pregnancy.
Enterogermina should be used during pregnancy alone if the benefits for the mother outweigh the risks, including those for the foetus.

Breast-feeding
No data are available on the use of Enterogermina during breastfeeding with regard to the composition of breast milk and the effects on the baby. It is not possible to draw conclusions on the safety of the use of Enterogermina during breastfeeding.
Enterogermina should be used during breastfeeding only if the potential benefits for the mother outweigh the potential risks, including those for the breastfed baby.

Fertility
No data are available on the effect of Enterogermina on human fertility.

Effects on ability to drive and use machines
The drug does not interfere with the ability to drive or use machines.

Undesired effects
During the treatment with this medicinal product the following undesirable effects were observed and classified according to MedDRA classification in organ classes and according to the following frequency classes:
Very common (≥1/10); Common (≥1/100,<1/10); Uncommon (≥1/1000,<1/100); Rare (≥1/10,000,<1/1000); Very rare <1/10,000); Unknown (the frequency cannot be defined based on available data).

Classification based on systems and organs	Common	Uncommon	Rare	Very rare	Unknown
Infections and infestations					Bacteraemia, septicaemia and sepsis (in immunocompromised or severely ill patients) (see section Special warning and Precaution for use)
Skin and subcutaneous disorders					hypersensitive reactions, including rash, hives and angioedema

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Reporting of suspected adverse reactions.
Reporting suspected adverse reactions after authorisation of the medical product is important. It allows continued monitoring of the benefit/risk balance of the medical product.

Overdose
No cases of overdose have been reported.

PHARMACOLOGICAL PROPERTIES
Pharmacodynamic properties
Pharmacotherapeutic category: A07FA – anti-diarrhoeal microorganisms
ENTEROGERMINA is a product consisting of a suspension of 4 strain (SIN, O/C, T, N/R) Bacillus clausii spores, which occur naturally in the intestine and are non-pathogenic.
When administered orally, the elevated resistance of Bacillus clausii spores to both chemical and physical agents allows them to cross the barrier of gastric juice, and to be unharmed when they reach the intestinal tract where they are transformed into metabolically active vegetative cells.
Spores can survive heat and gastric acidity, by nature. In a validated in vitro model Bacillus clausii spores demonstrated to survive in a simulated gastric environment (pH 1.4-1.5) until 120 minutes (survival rate of 96%). In a model that simulates the intestinal environment (saline solution of bile and pancreatin - pH 8), Bacillus clausii spores demonstrated their capability to multiply compared to the initial amount, in a statistically significant way (from 10^9 to 10^{12} CFU - Colony-Forming Units), starting from 240 minutes after the incubation. In a study that was conducted on 20 individuals, it was noticed that, in humans, Bacillus Clausii spores persist in the intestine and can be found in feces until 12 days after a single oral administration.
The administration of ENTEROGERMINA contributes to the restoration of intestinal microbial flora altered by dysmicrobism, also known as dysbiosis, that results from the antibiotic therapy and that can be associated with gastrointestinal symptoms, e.g. diarrhoea, abdominal pain and increase of air in the intestine.
In two open randomized controlled clinical trials, ENTEROGERMINA demonstrated to reduce the duration of acute diarrhoea in children older than 6 months. When taken during the antibiotic treatment and the next 7-10 days, ENTEROGERMINA demonstrated to reduce the incidence of abdominal pain and diarrhoea that are associated with the antibiotic treatment.

A prospective, observational, multicentre study conducted on 261 patients evaluated the 'real world' use of the probiotic Bacilus clausii for symptoms such as diarrhoea, pain, bloating, meteorism, constipation and abdominal tension through the administration of a questionnaire by the pharmacist before starting to take it and after 30 days.
Patients reported taking the medicinal product mainly for diarrhoea (56.7%), abdominal pain (13.41%) and bloating (12.64%). The mean duration of treatment was 7.1 days. On average, the improvement in symptoms was perceived after it had been taken for 3 days. After treatment, 95% of patients reported an improvemnent in diarrhoea and 97% an improvement in adbominal pain.
The 2 main mechanisms, reported below, contribute to Bacillus clausii effect of restoring the intestinal bacterial flora.

<u>Growth Inhibition of Pathogenic Bacteria</u>
The three B. clausii supposed mechanisms of action are: colonization of free ecological niches, that are made unavailable by the growth of other microorganisms; competition for the bond with epithelial cells, that is particularly relevant for the spores in the germination initial and intermediate phases; production of antibiotics and/or enzymes that are secreted in the intestinal environment.
In an in vitro study Bacillus clausii spores demonstrated to produce bacteriocins and antibiotics such as clausin, with antagonist activity against Gram-positive bacteria Staphylococcus aureus, Clostridium difficile, Enterococcus faecium.

<u>Immunomodulatory activity</u>
Bacillus clausii spores, administered through the oral route, in in vitro and in vivo murine models demonstrated to stimulate the production of Interferon-gamma and to increase the CD4+ T Lymphocyte proliferation.
Furthermore, *Bacillus clausii* demonstrated its capability of producing various B vitamins, aiding in correcting avitaminosis due to intestinal bacterial flora imbalance.
Furthermore, the high level of artificially induced heterologous resistance to antibiotics creates the therapeutic conditions for preventing the alteration of microbial intestinal flora by the selective action of antibiotics, particularly broad-spectrum antibiotics, or for restoring the intestinal microbial flora.
Due to its antibiotic resistance, ENTEROGERMINA may be administered between two subsequent administrations of antibiotics.
Antibiotic resistance refers to: penicillins, if not in combination with beta-lactamase inhibitors, cephalosporins (partial resistance in most cases), tetracyclines, macrolides, aminoglycosides (except for gentamicin and amikacin), chloramphenicol, thiamphenicol, lincomycin, clindamycin, isoniazid, cycloserine, novobiocin, rifampicin, nalidixic acid and pipemidic acid (intermediate resistance), and metronidazole.

Pharmacokinetic properties
<u>Absorption and elimination</u>
ENTEROGERMINA is not absorbed from the gastrointestinal tract. In animals, low numbers of B. clausii cells were found in mesenteric ganglia after administration of large doses of ENTEROGERMINA but no cells were found in the blood or spleen.
No intrinsic factor (age, sex, race, genetic polymorphism, renal or hepatic impairment) is likely to influence the fate of ENTEROGERMINA in the organism.

PHARMACEUTICAL PARTICULARS
List of excipients
Oral suspension: Purified water.

Shelf-life
As indicated on the outer package.

Special precautions for storage
Store at a temperature no higher than 30°C.

Nature and contents of container
Cardboard box containing 10 or 20 minibottles which made of LDPE.
Not all pack size are marketed.

Special precautions for disposal and handling
Oral suspension: shake the minibottle before use.

MANUFACTURER
Opella Healthcare Italy S.r.l. – Viale Europa 11, 21040 Origgio (VA), Italy

DATE OF REVISION OF THE TEXT
July 2025 (base on SmPC August 2024)

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